



A novel herbal treatment reduces depressive-like behaviors and increases BDNF levels in the brain of stressed mice

Ravid Doron^{a,b,c,*}, Dafna Lotan^{a,d,e}, Nili Einat^b, Roni Yaffe^a, Avigail Winer^a, Inbal Marom^a, Gili Meron^b, Nadav Katelny^f, Moshe Rehavi^{g,h}

^a School of Behavioral Sciences, Academic College of Tel Aviv-Yaffo, Tel-Aviv 61083, Israel

^b Department of Education and Psychology, The Open University of Israel, 108 Ravutski St., P.O. Box 808, Raanana 43107, Israel

^c School of Health and Life Sciences, Hadassah Academic College, Jerusalem 94702, Israel

^d School of Psychological Sciences, Tel-Aviv University, Tel-Aviv 69978, Israel

^e Sagol School of Neuroscience, Tel-Aviv University, Tel-Aviv 69978, Israel

^f Broshim Campus, Tel-Aviv University, Tel-Aviv, Israel

^g Department of Physiology and Pharmacology, Sackler Faculty of Medicine, Tel-Aviv University, Tel-Aviv 69978, Israel

^h The Dr. Miriam and Sheldon G. Adelson Chair in the Biology of Addictive Diseases, Sackler Faculty of Medicine, Tel-Aviv University, Tel-Aviv 69978, Israel

ARTICLE INFO

Article history:

Received 29 August 2013

Accepted 18 October 2013

Keywords:

BDNF

Depression

Herbal treatment

Mice

Serotonin transporter

Sexual dysfunction

SSRI

ABSTRACT

Aims: Depression is a chronic, recurring and potentially life-threatening illness. Current treatments for depression are characterized by a low success rate and associated with a wide variety of side effects. The aim of the present study was to evaluate the behavioral and biological anti-depressant effects of a novel herbal treatment (NHT), as well as to assess its potential side effects, in comparison to treatment with the selective serotonin reuptake inhibitor escitalopram.

Main methods: Depressive-like behavior was evaluated using the forced swim test (FST) and the tail suspension test (TST). Sexual behavior was evaluated following treatment by measuring latency before first mount and number of total mounts. Brain derived neurotrophic factor (BDNF) levels were evaluated using enzyme-linked immunosorbent assay. Serotonin transporter (SERT) levels in the pre-frontal cortex (PFC) and hypothalamus were evaluated using high affinity binding assay.

Key findings: (1) The NHT reduced depressive-like behavior in the FST and TST; (2) BDNF levels in the PFC of mice treated both with the NHT and escitalopram were increased; (3) SERT levels in the hypothalamus were significantly higher in the NHT group, in comparison to escitalopram and the control groups, and significantly lower in the PFC of the NHT group in comparison to the escitalopram group; and (4) the NHT led to less sexual dysfunction, compared to treatment with escitalopram.

Significance: Our NHT has the potential of being highly efficacious in treating depression in humans, while causing minimal to no influence on sexual function.

© 2013 Elsevier Inc. All rights reserved.

Introduction

Depression is a chronic and potentially life-threatening illness, with a life prevalence of 20–25% of women and 10–17% of men (Levinson, 2006). Recent work has indicated that stress exposure may interact with genetic risk factors to increase susceptibility to depression (Caspi et al., 2003; Kaufman et al., 2006). The relation between stress and risk for depression has been documented for both acute (Kendler et al., 1998) and chronic stress (Hammen et al., 2009).

Finding adequate treatments for depression is of the utmost importance. There are several pharmacotherapeutic approaches in the treatment of depression, including tricyclic drugs, monoamine oxidase

inhibitors, serotonin-norepinephrine reuptake inhibitors and selective serotonin reuptake inhibitors (SSRIs). Nevertheless, these are characterized by a low success rate and associated with a wide variety of side effects (Lam and Kennedy, 2004).

The disadvantages of conventional pharmacological treatment have prompted the search for alternative treatments for depression. One such treatment is the use of herbal medicines. Studies have shown therapeutic potential of chronic treatment with several herbal medicines, such as St. John's Wort (Linde et al., 2008) and Kava (Witte et al., 2005), in various psychiatric disorders. Chronic treatment with such herbal medicines was shown to have antidepressant-like effect, to normalize brain transmitter levels (Simmen et al., 1999) and to increase brain derived neurotrophic factor (BDNF) levels (Hou et al., 2010; Li et al., 2009), similar to the effects seen following SSRI treatment. The exact mechanisms of action of the herbal compounds are still unclear, but the available research clearly shows that various bioactive

* Corresponding author at: School of Behavioral Sciences, Academic College of Tel Aviv-Yaffo, Tel-Aviv 61083, Israel. Tel.: +972 3 6209703; fax: +972 3 6802526.

E-mail address: raviddor@mta.ac.il (R. Doron).

constituents contribute to the clinical effects reported, often in a synergistic manner and with minimal or nonexistent side effects.

The present study has focused on two main goals: (a) evaluating the behavioral and biological anti-depressant effects of a novel herbal treatment (NHT) (U.S. PCT No. 61/311,537), in comparison to the conventional SSRI treatment, namely, escitalopram, and (b) assessing one of the main side effects of the conventional treatment, namely, sexual dysfunction, as a result of treatment with NHT in comparison to treatment with escitalopram. The NHT was prepared from the following four components: *Crataegus pinnatifida*, *Triticum aestivum*, *Lilium brownie* and *Fructus zizyphi jujubae*, as a modification of a classical Chinese formula used in the treatment of mental disorders since the 2nd century A.D. (Scheid et al., 2009). Previous studies have pointed at the behavioral and biological anti-depressant effects of the NHT components (Butterweck et al., 2001; Chen et al., 2012; Lin et al., 2003; Seely and Singh, 2007; Zhu et al., 2006). We (Doron et al., 2012) have previously found that this treatment displayed behavioral and biological anxiolytic effects in treated mice, compared to control mice and similar to the ones found following escitalopram treatment. In the present study we evaluated depressive-like behavior in adult mice exposed to stress during adolescence following three weeks of treatment with the NHT or escitalopram. The behavioral evaluation in these mice was followed by analysis of BDNF and serotonin transporter (SERT) levels in the brain. In addition, following three weeks of treatment with either the NHT or escitalopram, naïve and stressed mice were tested for one of the main side effects of SSRI treatment: sexual dysfunction.

Materials and methods

Drugs

Crataegus pinnatifida, *Triticum aestivum*, *Lilium brownie* and *Fructus zizyphi jujubae* were purchased as freeze-dried granules from KPC Products, Inc. (Irvine, CA, USA). Escitalopram was kindly donated by TEVA Ltd. (Israel). The NHT was prepared by dissolving the 4 compounds (together) in saline containing 1% DMSO to give a final concentration of 0.47 mg/ml (each). The NHT was administered at a daily dose of 30 mg/kg, and escitalopram was administered at a daily dose of 15 mg/kg (i.p. injection).

Animals

For all experiments, ICR outbred mice (Harlan, Israel) were bred and kept in the vivarium of the 'Academic College of Tel Aviv-Yafo'. Mice were housed in standard group cages (5 mice per cage, each cage contains mice from all experimental groups), kept on a reversed 12 h light/dark cycle (lights on 1900–0700) and given ad libitum access to food and water. All experiments were performed during the dark phase under red light. All experiments were approved by the International Committee for Animal Care and Use in Israel.

Biochemical assessments

Assessment of brain BDNF levels

Tissue samples were obtained as previously described (Maayan et al., 2006). Shortly, mice were decapitated, and their brains were placed on ice. Serial sections were cut onto slides and tissue punches of the hippocampus and PFC were taken. Tissue punches were homogenized in cold extraction buffer (Tris-buffered saline, pH 8.0, with 1% NP-40, 10% glycerol, 5 mM sodium metavanadate, 10 mM PMSF, 100 µg/ml aprotinin and 10 µg/ml leupeptin). Homogenates were acidified with 0.1 M HCl (pH 3.0), incubated at room temperature (22–24 °C) for 15 min, and neutralized (pH 7.6) with 0.1 M NaOH. Homogenates were then microfuged at 7000 g for 10 min. BDNF levels were evaluated using sandwich enzyme-linked immunosorbent assay as previously described (Baker-Herman et al., 2004).

Assessment of brain SERT levels

Levels of brain SERT were evaluated using high affinity [³H] citalopram binding assays. Shortly, mice were decapitated and their brains were dissected on ice. Mice brain PFC and hypothalamus were disrupted with Brinkman polytron in 50 vol of buffer (50 mM Tris-HCl, 120 mM NaCl and 5 mM KCl at a pH of 7.4) and centrifuged ($\times 3$) at 30,000 $\times g$ for 10 min. The pellet was resuspended in the same buffer to yield a final concentration of about 21 mg/ml (wet weight). [³H]Citalopram binding was determined by a standard binding assay that contained 100 µl of brain homogenate, 100 µl [³H]citalopram (0.54 nM) and 300 µl buffer. After a 60 min incubation period at 25 °C, homogenates were diluted in 3 ml ice-cold buffer and filtered with vacuum through Whatman GF/C glass fiber filters. Filters were washed ($\times 3$) with 3 ml ice-cold buffer, and the radioactivity was measured in scintillation liquid using a β -counter (Packard, Tri-Carb 2100TR). Specific binding was defined as the difference between total [³H]citalopram binding (triplicate samples) and the binding in the presence of 10 µM fluoxetine (duplicate samples). Protein concentration was measured by the method of Lowry et al. (1951).

Stress procedures

Unpredictable chronic mild stress (UCMS)

This paradigm is based on the procedure originally designed by Willner et al. (1992) and adapted to mice (Ducotet and Belzung, 2004). The procedure was performed during adolescence, starting at the age of 4 weeks. Treated mice were subjected to UCMS for 4 weeks, using the following stressors: placement in an empty cage with 1 cm of water at the bottom (water stress), inducing social stress by placing mice in soiled cages of other mice, inverting the light/dark cycle, placing mice in cages with wet sawdust, tilt cages at 30 degrees and restraining the mice. To prevent habituation and to provide an unpredictable feature to the stressors, stressors were administered at different time points during the day (see Table 1).

Behavioral assessment

Forced swim test (FST)

The FST was used to monitor depression-like behavior. This paradigm is based upon the evaluation of immobility, as a measure of behavioral despair (Cryan and Holmes, 2005). Shortly, after being placed in a transparent plexiglas cylinder (20 cm diameter) filled with water (25 °C) to a depth of 12 cm, mice were video recorded for 6 min, and later coded by the Biobserve software. Total time spent immobile during the last 4 min was measured. Immobility was defined as the cessation of limb movements except minor movement necessary to keep the mouse afloat.

Tail suspension test (TST)

Mice behavioral despair was evaluated in the TST by measuring immobility (Steru et al., 1987). Mice were suspended from a horizontal bar by taping the tip of their tail to the bar for 6 min, and the time spent in immobile positions during the last 4 min was evaluated by Biobserve software.

Evaluation of sexual dysfunction

The effect of the NHT on sexual dysfunction was evaluated in 2 mice groups: naïve mice and mice exposed to UCMS. Each of these groups was subsequently divided to the 3 groups of treatment. The influence of the NHT on sexual behavior was tested following a 3-week treatment. A male mouse was placed with a female mouse in estrus, in the male's home cage during the dark phase under red dim light for 30 min. Male sexual behaviors, i.e., latency before first mount and number of total mounts, were recorded and later coded by an observer blind to the mouse treatment.

Table 1

Unpredictable chronic mild stress (UCMS) procedure. Treated mice were subjected to UCMS for 4 weeks, using the listed stressors. To prevent habituation and to provide an unpredictable feature to the stressors, stressors were administered at different time points during the day.

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Week 1	Tilt cages at 30 degrees (4 h)	Placement in an empty cage with water at the bottom + lights on (4 h)	Tilt cages at 30 degrees (4 h)	Placing mice in cages with wet sawdust (4 h)	Placing mice in soiled cages of other mice (4 h)	Restraint stress (4 h)	Reversal of the light/dark cycle
Week 2	Placement in an empty cage with water at the bottom + Lights on (4 h)	Tilt cages at 30 degrees (4 h)	Placing mice in cages with wet sawdust (4 h)	Placing mice in soiled cages of other mice (4 h)	Restraint stress (4 h)	Placement in an empty cage with water at the bottom + Lights on (4 h)	Reversal of the light/dark cycle
Week 3	Tilt cages at 30 degrees (4 h)	Placing mice in cages with wet sawdust (4 h)	Placing mice in soiled cages of other mice (4 h)	Restraint stress (4 h)	Placement in an empty cage with water at the bottom + lights on (4 h)	Tilt cages at 30 degrees (4 h)	Reversal of the light/dark cycle
Week 4	Placing mice in cages with wet sawdust (4 h)	Restraint stress (4 h)	Tilt cages at 30 degrees (4 h)	Restraint stress (4 h)	Placement in an empty cage with water at the bottom + lights on (4 h)	Placing mice in cages with wet sawdust (4 h)	Reversal of the light/dark cycle

Data analysis and interpretation of results

Results are expressed as mean $\pm$ SEM. Data was analyzed using a one-way ANOVA with treatment as a between subject variable. For the FST and TST procedures, where results were significant, the ANOVA was followed by a planned contrast analysis. Significance was assumed as $p < 0.05$. For the brain BDNF levels; brain SERT levels; time until first mount; and the number of total mounts, where results were significant, the ANOVA was followed by a post hoc LSD analysis.

Results

The NHT has led to an increase in Brain BDNF levels

One way-ANOVA revealed significant main effect of treatment on BDNF levels in the PFC ($F(2,12) = 4.536$, $p < 0.05$, Fig. 1). Post hoc analysis has revealed that mice treated for 3 weeks with the NHT (Fig. 1, $p < 0.05$) or escitalopram (Fig. 1, $p < 0.05$) had significantly higher BDNF levels in the PFC in comparison to control mice (Fig. 1, $p < 0.05$). No differences in hippocampus BDNF levels were found between the different treatment groups ($F(2, 53) = 0.841$, $p = 0.44$).

Different treatments have yielded different changes in brain SERT levels

One way-ANOVA revealed significant main effect of treatment on SERT levels in the hypothalamus ($F(2,10) = 8.047$, $p < 0.01$, Fig. 2A). Post hoc analysis has revealed that SERT levels in the hypothalamus of mice treated for 3 weeks with the NHT were significantly higher compared to both the control (Fig. 2A, $p < 0.005$) and escitalopram (Fig. 2A, $p < 0.05$) treated mice. Nevertheless, one way-ANOVA revealed significant main effect of treatment on SERT levels in the PFC ($F(2,32) = 32.555$, $p < 0.001$, Fig. 2B). Post hoc analysis has revealed that SERT levels in the PFC of escitalopram treated mice were significantly lower compared to both the control (Fig. 2B, $p < 0.005$) and NHT mice (Fig. 2B, $p < 0.05$).

The NHT reduced depressive-like behavior

Depressive-like behavior was evaluated in treated mice previously exposed to a protocol of UCMS in adolescence. Upon reaching adulthood and following 3 weeks of treatment, depressive-like behavior was assessed in the FST and the TST. Treatment has reduced depressive-like behavior both in the FST ($F(2,43) = 5.003$, $p < 0.05$, Fig. 3A) and the TST ($F(2,45) = 3.313$, $p < 0.05$, Fig. 3B). Planned contrast analyses have revealed that treatment with either escitalopram or the NHT has reduced depressive-like behavior in the FST (Fig. 3A, $p < 0.05$ for both treatments) and TST (Fig. 3B, $p < 0.05$ for both treatments), in comparison to control mice.

The NHT led to less sexual dysfunction compared to the conventional treatment

Mice sexual behavior was assessed following the different treatments, by evaluating the time before the first mount and total number of mounts. One way-ANOVA revealed significant main effect of treatment on number of mounts ($F(2,55) = 4.542$, $p < 0.05$, Fig. 4A). Post hoc analysis has revealed that naïve mice, treated for 3 weeks with escitalopram, displayed significantly lower number of mounts compared to control mice (Fig. 4A, $p < 0.05$). There were no significant differences in this parameter between the NHT and control groups. There were no significant differences in time before first mount between the different treatment groups ($F(2, 57) = 1.867$, $p = 0.164$, Fig. 4B). In mice exposed to UCMS, one way-ANOVA revealed significant main effects of treatment on the number of mounts ($F(2,54) = 8.635$, $p < 0.001$, Fig. 4C). Post hoc analysis has revealed that mice treated with escitalopram displayed significantly lower number of mounts compared to both the NHT (Fig. 4C, $p < 0.05$) and control mice (Fig. 4C, $p < 0.001$). In addition, there was a significant main effect of treatment on the time before the first mount in UCMS exposed mice ($F(2,57) = 4.114$, $p < 0.05$, Fig. 4D). Post hoc analysis has revealed that mice treated with escitalopram, had spent significantly longer time before the first mount compared to control mice (Fig. 4D, $p < 0.05$). There was a tendency to significance in this parameter between the escitalopram and the NHT groups (Fig. 4D, $p = 0.087$).

Discussion

The present study explored the anti-depressant properties of a NHT (U.S. PCT No. 61/311,537), as well as its potential effect on sexual behavior, in comparison with the conventional escitalopram treatment. The study led to four main findings: (1) the NHT reduced depressive-like behavior following unpredictable stress during adolescence in two behavioral tests, namely, the forced swim test (FST) and the tail suspension test (TST). (2) BDNF levels in the PFC of mice treated with the NHT or with escitalopram were increased; (3) SERT levels in the hypothalamus were significantly higher in the NHT group, in comparison to escitalopram treated group and the control group, while these levels were significantly lower in the PFC of the NHT mice in comparison to escitalopram treated mice, but were not different from the control; and (4) the NHT has led to less sexual dysfunction, compared to treatment with escitalopram.

We have recently found (Doron et al., 2012) that the NHT has reduced anxiety-like behavior in mice previously exposed to stress. In order to explore the behavioral anti-depressant aspect, we have employed two of the most commonly used animal models of depression for antidepressant screening, namely, the FST and TST. The FST model has a high degree of pharmacological validity as evidenced by its sensitivity to major classes of antidepressants (Borsini and Meli, 1988), and

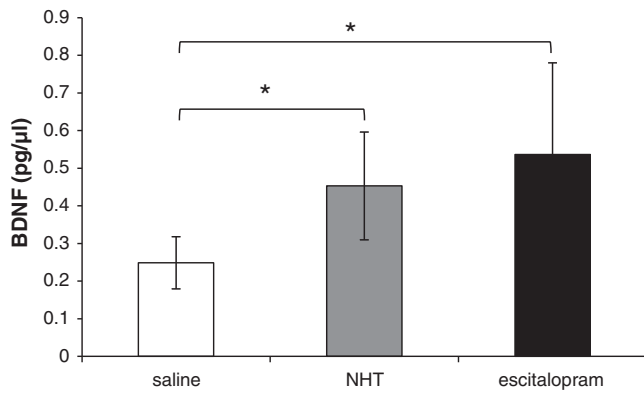


Fig. 1. Effect of the novel treatment (NHT) and escitalopram treatment (3 weeks) on brain derived neurotrophic factor (BDNF) levels in the pre-frontal cortex (PFC) (N: NHT = 5, escitalopram = 5, control = 5). * $p < 0.05$.

the immobility observed in this test reflects certain aspects of depression and is reduced by a variety of antidepressants (David et al., 2001; Redrobe et al., 1998). The FST may require additional tests to distinguish antidepressants from some other drug classes. Therefore, the TST has been used. The TST has been shown to be sensitive to various antidepressants, and avoids problems of hypothermia or motor dysfunction that could interfere with performance in the FST (Steru et al., 1985). The NHT has shown reduced immobility time in both the FST and TST, in comparison to control mice. Interestingly, conventional treatment with escitalopram has gained a similar anti-depressant effect. Previous studies have shown that herbal medicines reduced immobility in both FST and TST, in a similar way to the treatment with different SSRIs, such as imipramine and fluoxetine both in mice and in rats (Wei et al., 2008; Zhu et al., 2006). These findings are consistent with clinical trials, in which treatment with different herbal compounds, such as St. John's Wort, Saffron, Rhodiola and Lavender had anti-depressant effects in depressed patients, and yield similar effects to the conventional treatments, such as fluoxetine and imipramine (Akhondzadeh et al., 2003; Dwyer et al., 2011).

Another interesting finding was that the NHT has increased the hypothalamic serotonin transporter (SERT) levels, while treatment with escitalopram did not change SERT levels in comparison to controls. Interestingly, SERT levels in the PFC of escitalopram treated mice were significantly lower, compared to both the NHT and control mice. Various studies have demonstrated that SERT plays a pivotal role in depression. Polymorphism in the SERT gene has been repeatedly correlated with depression, especially for patients with a history of severe life events (Caspi et al., 2003). In addition, imaging studies have shown a reduction in SERT levels in the hypothalamus and PFC of major depression

patients (Joensuu et al., 2010; Malison et al., 1998), while other studies have shown that SERT levels in depressed patients are elevated and that this finding can already be observed from childhood (Dahlstrom et al., 2000). Treatments with SSRIs have been shown to reduce SERT levels in different brain regions, such as the hypothalamus and PFC (Dahlstrom et al., 2000; Williams et al., 2005). Nevertheless, our NHT has significantly increased SERT levels in the hypothalamus, and did not change its levels in the PFC, compared to control mice. The contradiction between the influence of escitalopram and our NHT on SERT levels can be explained in a few ways. First, SERT levels might have an inverted-U shape influence on depression. Though high levels of SERT can induce depression, very low levels of SERT can also induce this situation, as can be seen in SERT knockout mice. This means that either under activity or over activity of SERT can cause depression (Calabrese and Baldwin, 2001; Olivier et al., 2008). Second, this finding might point to a different mechanism by which the NHT improves the depression-like behavior.

One of the most important findings of our study was that treatments with either the NHT or escitalopram have increased BDNF levels in the PFC of treated mice. Brain derived neurotrophic factor (BDNF) is a critical regulator of the formation, plasticity, and integrity of neurons in brain circuits that regulate emotion (Duman and Monteggia, 2006). The “neurotrophin hypothesis” postulates that reduced activity of BDNF causes a depressive state (Chourbaji et al., 2011). Several studies have found decreased central (Karege et al., 2005) and peripheral levels of BDNF (Molendijk et al., 2011) in patients with major depression disorder. Human genetic studies have reported an association between certain BDNF genes and major depression disorder (Licinio et al., 2009; Schumacher et al., 2005). Other studies have found reduced mRNA levels of BDNF both in the prefrontal cortex and hippocampus of suicide subjects, as compared with those in control subjects (Dwivedi et al., 2003). Consistent with these findings, studies in mice have shown that BDNF deletion mutants also display a depressive phenotype when exposed to mild stress (Duman and Monteggia, 2006). Chronic SSRI treatment has been repeatedly shown to increase BDNF levels in the serum of treated patients (Brunoni et al., 2008; Sen et al., 2008), and in the brains of treated rodents (Delahanty and Nugent, 2006). Increased levels of BDNF in the PFC of the NHT and SSRI treated groups are consistent with the human and animal studies mentioned above.

One of the most encountered side effects reported following long-term treatment with SSRIs is sexual dysfunction, and is the most frequent reasons cited by patients for SSRI discontinuation (Cascade et al., 2009). Approximately 14–59% of patients receiving SSRI treatment report on having sexual dysfunction (Montejo et al., 2001). Treatment with different SSRIs has been shown to cause delayed or absence of ejaculations, anorgasmia and decreased libido in depressed patients (Chan et al., 2011; Soga et al., 2010). In accordance with these clinical

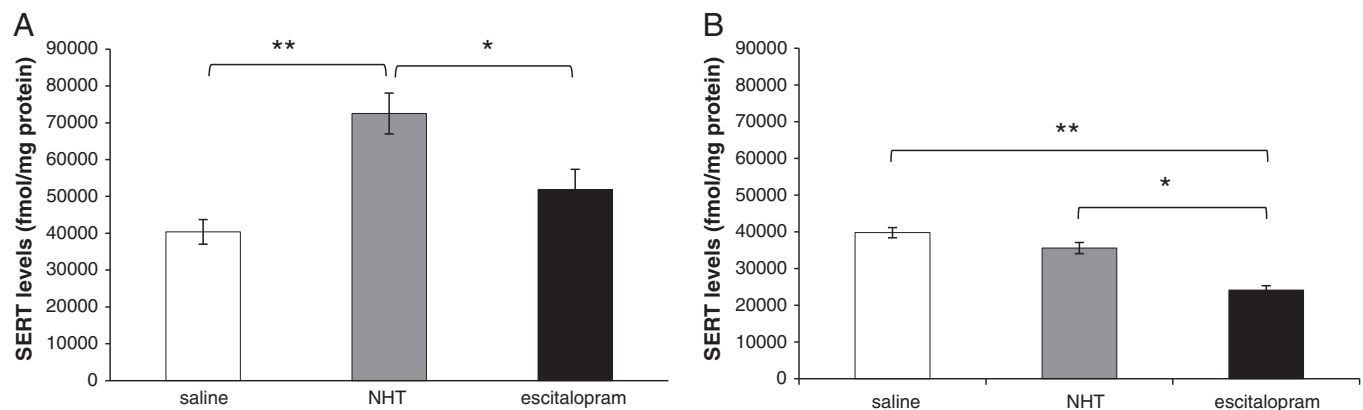


Fig. 2. Effect of the NHT and escitalopram treatment (3 weeks) on serotonin transporter (SERT) levels (at 0.54 nM [3 H]citalopram) (in the hypothalamus and pre-frontal cortex (PFC)). A: Treatment effect on hypothalamus SERT levels (N: NHT = 5, escitalopram = 4, control = 4). B: Treatment effect on PFC SERT levels (N: NHT = 10, escitalopram = 10, control = 15). * $p < 0.05$, ** $p < 0.005$.

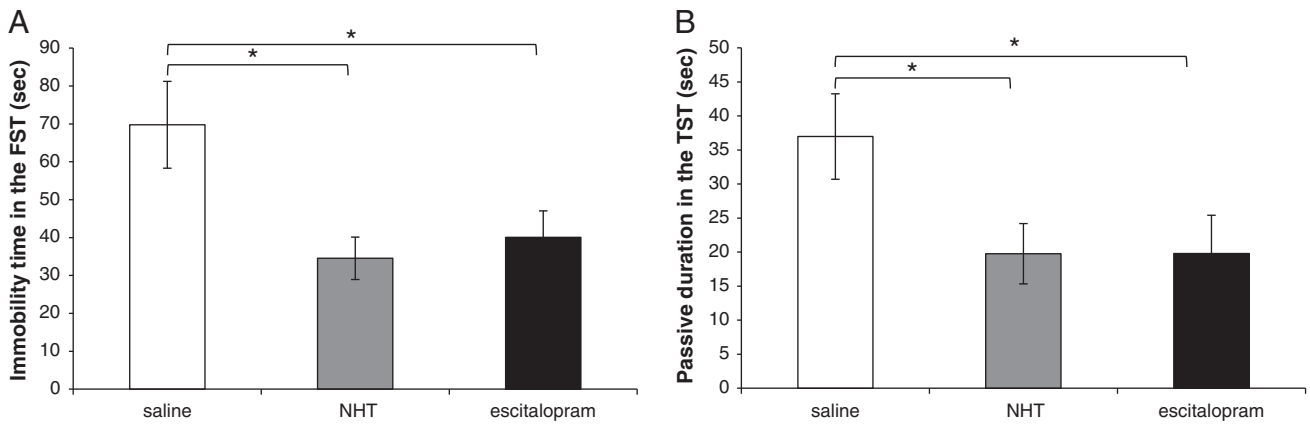


Fig. 3. Effect of the NHT and escitalopram treatment (3 weeks) on depressive-like behavior. A: Effect of treatment on immobility time in the forced swim test (FST) (N: NHT = 15, escitalopram = 15, control = 16). B: Effect of treatment on passive duration in the tail suspension test (TST) (N: NHT = 17, escitalopram = 15, control = 16). * $p < 0.05$.

findings, we have found that treatment with escitalopram led to a significant sexual dysfunction as can be seen in a longer latency before the first mount and lower number of mounts in treated mice, in comparison to both the NHT and control mice. This effect was more profound in UCMS exposed mice than in naïve mice. Our results are consistent with previous animal studies that demonstrated a significant sexual dysfunction following chronic SSRI treatments, as measured by the latency of the first mount, number of mounts with and without intromission, the number of contacts and post ejaculatory interval (de Jong et al., 2005; Soga et al., 2010). Our NHT has not led to sexual dysfunction in treated mice. These finding might point at a different mechanism, by which the NHT yields its beneficial anti-depressant effect, while not altering the deleterious side effects observed following

the conventional treatment. These findings might be explained by our results showing differences in SERT levels following the different treatments. Sexual dysfunction has been shown to be a result of changes in levels of SERT and serotonin in different brain regions. An elevation in serotonin levels in the PFC of citalopram treated mice, which is probably a result of a reduction in SERT levels in this brain region, was correlated with impaired sexual behavior of treated mice (de Jong et al., 2006). Chan et al. (2011) have found that SERT homozygote and heterozygote knock out rats have displayed altered sexual behavior. These rats had a reduction in SERT levels, which is similar to the one observed following chronic SSRI treatment. The differences in SERT levels between escitalopram and the NHT groups are in line with the literature findings, and can explain the significant appearance in side effects in the

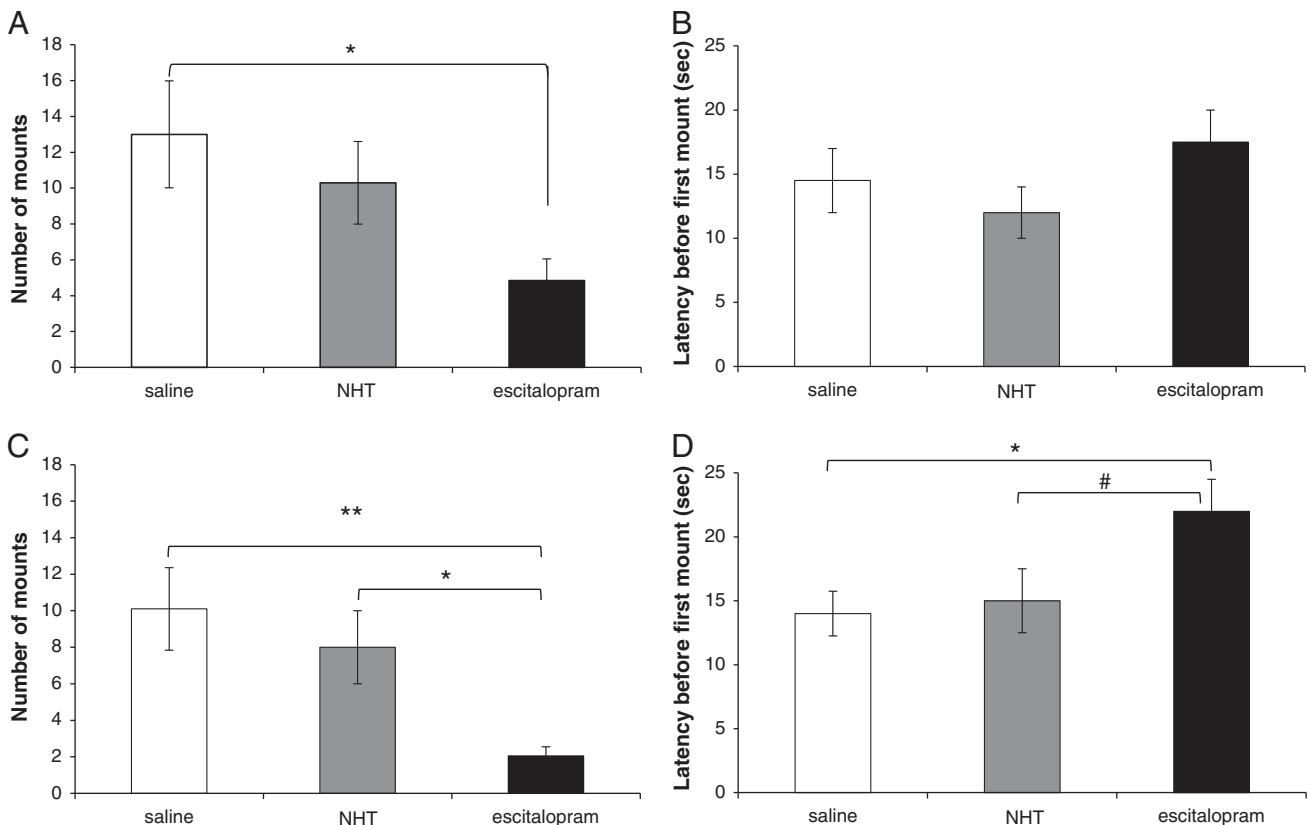


Fig. 4. Effect of the NHT and escitalopram treatment (3 weeks) on sexual behavior. A: Effect of treatment on number of mounts in naïve mice (N: NHT = 20, escitalopram = 19, control = 19). B: Effect of treatment on latency before the first mount in naïve mice (N: NHT = 21, escitalopram = 20, control = 19). C: Effect of treatment on number of mounts in mice previously exposed to unpredictable chronic mild stress (UCMS) (N: NHT = 18, escitalopram = 19, control = 20). D: Effect of treatment on latency before the first mount in mice previously exposed to UCMS (N: NHT = 20, escitalopram = 20, control = 20). * $p < 0.05$, ** $p < 0.001$, # $p = 0.087$.

escitalopram treated mice and absence in side effects following the NHT.

Conclusions

This study has demonstrated the potential anti-depressant effects of a NHT (U.S. PCT No. 61/311,537). The precise mechanism(s) underlying the effects of our NHT remain to be determined but are likely to involve, in part, changes in BDNF and SERT levels in the brain. These behavioral and biochemical findings indicate that our NHT has the potential of being highly efficacious in treating depression, while causing minimal to no sexual dysfunction.

Conflict of interest statement

Dr. Ravid Doron and Mr. Nadav Kately have an approved patent relating to the herbal treatment for anxiety disorders (PCT 61-311,537, USA). All authors assert that none has any commercial or financial involvements that might present an appearance of a conflict of interest in connection with the submitted manuscript.

Funding

This work was funded by the Israel Science Foundation (ISF 738/11), by the National Institute for Psychobiology in Israel (NIPI-7-2011-12), and by the Open University Foundation. All authors assert that none has any commercial or financial involvements that might present an appearance of a conflict of interest in connection with the submitted manuscript.

Acknowledgments

We are especially grateful to Mrs. Neta Roz, for her assistance in the assessment of brain SERT levels. We would also like to thank Dr. Xiao Shi for his contribution of clinical knowledge.

References

- Akhondzadeh S, Faraji H, Sadeghi M, Afkham K, Fakhrazadeh H, Kamalipour A. Double-blind comparison of fluoxetine and nortriptyline in the treatment of moderate to severe major depression. *J Clin Pharm Ther* 2003;28:379–84.
- Baker-Herman TL, Fuller DD, Bavis RW, Zabka AG, Golder FJ, Doperalski NJ, et al. BDNF is necessary and sufficient for spinal respiratory plasticity following intermittent hypoxia. *Nat Neurosci* 2004;7:48–55.
- Borsini F, Meli A. Is the forced swimming test a suitable model for revealing antidepressant activity? *Psychopharmacology (Berl)* 1988;94:147–60.
- Brunoni AR, Lopes M, Fregni F. A systematic review and meta-analysis of clinical studies on major depression and BDNF levels: implications for the role of neuroplasticity in depression. *Int J Neuropsychopharmacol* 2008;11:1169–80.
- Butterweck V, Nishibe S, Sasaki T, Uchida M. Antidepressant effects of apocynin venetum leaves in a forced swimming test. *Biol Pharm Bull* 2001;24:848–51.
- Calabrese EJ, Baldwin LA. U-shaped dose–responses in biology, toxicology, and public health. *Annu Rev Public Health* 2001;22:15–33.
- Cascade E, Kalali AH, Kennedy SH. Real-world data on SSRI antidepressant side effects. *Psychiatry (Edmont)* 2009;6:16–8.
- Caspi A, Sugden K, Moffitt TE, Taylor A, Craig IW, Harrington H, et al. Influence of life stress on depression: moderation by a polymorphism in the 5-HTT gene. *Science* 2003;301:386–9.
- Chan JS, Snoeren EM, Cuppen E, Waldinger MD, Olivier B, Oosting RS. The serotonin transporter plays an important role in male sexual behavior: a study in serotonin transporter knockout rats. *J Sex Med* 2011;8:97–108.
- Chen ML, Gao J, He XR, Chen Q. Involvement of the cerebral monoamine neurotransmitters system in antidepressant-like effects of a Chinese herbal decoction, Baihe Dihuang Tang, in mice model. *Evid Based Complement Alternat Med* 2012;2012:419257.
- Chourbaji S, Brandwein C, Gass P. Altering BDNF expression by genetics and/or environment: impact for emotional and depression-like behaviour in laboratory mice. *Neurosci Biobehav Rev* 2011;35:599–611.
- Cryan JF, Holmes A. The ascent of mouse: advances in modelling human depression and anxiety. *Nat Rev Drug Discov* 2005;4:775–90.
- Dahlstrom M, Ahonen A, Ebeling H, Tornaiainen P, Heikkila J, Moilanen I. Elevated hypothalamic/midbrain serotonin (monoamine) transporter availability in depressive drug-naïve children and adolescents. *Mol Psychiatry* 2000;5:514–22.
- David DJ, Bourin M, Hascoet M, Colombel MC, Baker GB, Jolliet P. Comparison of antidepressant activity in 4- and 40-week-old male mice in the forced swimming test: involvement of 5-HT1A and 5-HT1B receptors in old mice. *Psychopharmacology (Berl)* 2001;153:443–9.
- de Jong TR, Pattij T, Veening JG, Dederen PJ, Waldinger MD, Cools AR, et al. Citalopram combined with WAY 100635 inhibits ejaculation and ejaculation-related Fos immunoreactivity. *Eur J Pharmacol* 2005;509:49–59.
- de Jong TR, Snapaen LJ, Pattij T, Veening JG, Waldinger MD, Cools AR, et al. Effects of chronic treatment with fluvoxamine and paroxetine during adolescence on serotonin-related behavior in adult male rats. *Eur Neuropsychopharmacol* 2006;16:39–48.
- Delahanty DL, Nugent NR. Predicting PTSD prospectively based on prior trauma history and immediate biological responses. *Ann N Y Acad Sci* 2006;1071:27–40.
- Doron R, Lotan D, Rak-Rabl A, Raskin-Ramot A, Lavi K, Rehavi M. Anxiolytic effects of a novel herbal treatment in mice models of anxiety. *Life Sci* 2012;90:995–1000.
- Ducottet C, Belzung C. Behaviour in the elevated plus-maze predicts coping after subchronic mild stress in mice. *Physiol Behav* 2004;81:417–26.
- Duman RS, Monteggia LM. A neurotrophic model for stress-related mood disorders. *Biol Psychiatry* 2006;59:1116–27.
- Dwivedi Y, Rizavi HS, Conley RR, Roberts RC, Tamminga CA, Pandey GN. Altered gene expression of brain-derived neurotrophic factor and receptor tyrosine kinase B in postmortem brain of suicide subjects. *Arch Gen Psychiatry* 2003;60:804–15.
- Dwyer AV, Whitten DL, Hawrelak JA. Herbal medicines, other than St. John's Wort, in the treatment of depression: a systematic review. *Altern Med Rev* 2011;16:40–9.
- Hammen C, Kim EY, Eberhart NK, Brennan PA. Chronic and acute stress and the prediction of major depression in women. *Depress Anxiety* 2009;26:718–23.
- Hou Y, Aboukhatwa MA, Lei DL, Manaye K, Khan I, Luo Y. Anti-depressant natural flavonols modulate BDNF and beta amyloid in neurons and hippocampus of double TgAD mice. *Neuropharmacology* 2010;58:911–20.
- Joensuu M, Lehto SM, Tolmunen T, Saarinen PI, Valkonen-Korhonen M, Vanninen R, et al. Serotonin-transporter-linked promoter region polymorphism and serotonin transporter binding in drug-naïve patients with major depression. *Psychiatry Clin Neurosci* 2010;64:387–93.
- Karege F, Bondolfi G, Gervasoni N, Schwald M, Aubry JM, Bertschy G. Low brain-derived neurotrophic factor (BDNF) levels in serum of depressed patients probably results from lowered platelet BDNF release unrelated to platelet reactivity. *Biol Psychiatry* 2005;57:1068–72.
- Kaufman J, Yang BZ, Douglas-Palumberi H, Grasso D, Lipschitz D, Houshyar S, et al. Brain-derived neurotrophic factor-5-HTTLPR gene interactions and environmental modifiers of depression in children. *Biol Psychiatry* 2006;59:673–80.
- Kendler KS, Karkowski LM, Prescott CA. Stressful life events and major depression: risk period, long-term contextual threat, and diagnostic specificity. *J Nerv Ment Dis* 1998;186:661–9.
- Lam RW, Kennedy SH. Evidence-based strategies for achieving and sustaining full remission in depression: focus on metaanalyses. *Can J Psychiatry* 2004;49:175–265.
- Levinson DF. The genetics of depression: a review. *Biol Psychiatry* 2006;60:84–92.
- Li Q, Zhao HF, Zhang ZF, Liu ZG, Pei XR, Wang JB, et al. Long-term administration of green tea catechins prevents age-related spatial learning and memory decline in C57BL/6j mice by regulating hippocampal cyclic amp-response element binding protein signaling cascade. *Neuroscience* 2009;159:1208–15.
- Licinio J, Dong C, Wong ML. Novel sequence variations in the brain-derived neurotrophic factor gene and association with major depression and antidepressant treatment response. *Arch Gen Psychiatry* 2009;66:488–97.
- Lin RD, Hou WC, Yen KY, Lee MH. Inhibition of monoamine oxidase B (MAO-B) by Chinese herbal medicines. *Phytomedicine* 2003;10:650–6.
- Linde K, Berner MM, Kriston L. St John's wort for major depression. *Cochrane Database Syst Rev* 2008;CD000448.
- Lowry OH, Rosebrough NJ, Farr AL, Randall RJ. Protein measurement with the Folin phenol reagent. *J Biol Chem* 1951;193:265–75.
- Maayan R, Lotan S, Doron R, Shabat-Simon M, Gispán-Herman I, Weizman A, et al. Dehydroepiandrosterone (DHEA) attenuates cocaine-seeking behavior in the self-administration model in rats. *Eur Neuropsychopharmacol* 2006;16:329–39.
- Malison RT, Price LH, Berman R, van Dyck CH, Pelton GH, Carpenter L, et al. Reduced brain serotonin transporter availability in major depression as measured by [123I]-2 beta-carbomethoxy-3 beta-(4-iodophenyl)tropane and single photon emission computed tomography. *Biol Psychiatry* 1998;44:1090–8.
- Molendijk ML, Bus BA, Spinhoven P, Penninx BW, Kenis G, Prickaerts J, et al. Serum levels of brain-derived neurotrophic factor in major depressive disorder: state-trait issues, clinical features and pharmacological treatment. *Mol Psychiatry* 2011;16:1088–95.
- Montejo AL, Llorca G, Izquierdo JA, Rico-Villademoros F. Incidence of sexual dysfunction associated with antidepressant agents: a prospective multicenter study of 1022 outpatients. Spanish Working Group for the Study of Psychotropic-Related Sexual Dysfunction. *J Clin Psychiatry* 2001;62(Suppl. 3):10–21.
- Olivier JD, Van Der Hart MG, Van Swelm RP, Dederen PJ, Homberg JR, Cremers T, et al. A study in male and female 5-HT transporter knockout rats: an animal model for anxiety and depression disorders. *Neuroscience* 2008;152:573–84.
- Redrobe JP, Bourin M, Colombel MC, Baker GB. Psychopharmacological profile of the selective serotonin reuptake inhibitor, paroxetine: implication of noradrenergic and serotonergic mechanisms. *J Psychopharmacol* 1998;12:348–55.
- Scheid V, Bensky D, Ellis A, Barolet R. Chinese herbal medicine: formulas and strategies. 2nd ed. CA, USA: Eastland Pr; 2009.
- Schumacher J, Jamra RA, Becker T, Ohlraun S, Klopp N, Binder EB, et al. Evidence for a relationship between genetic variants at the brain-derived neurotrophic factor (BDNF) locus and major depression. *Biol Psychiatry* 2005;58:307–14.
- Seely D, Singh R. Adaptogenic potential of a polyherbal natural health product: report on a longitudinal clinical trial. *Evid Based Complement Alternat Med* 2007;4:375–80.
- Sen S, Duman R, Sanacora G. Serum brain-derived neurotrophic factor, depression, and antidepressant medications: meta-analyses and implications. *Biol Psychiatry* 2008;64:527–32.

- Simmen U, Burkard W, Berger K, Schaffner W, Lundstrom K. Extracts and constituents of *Hypericum perforatum* inhibit the binding of various ligands to recombinant receptors expressed with the Semliki Forest virus system. *J Recept Signal Transduct Res* 1999;19:59–74.
- Soga T, Wong DW, Clarke IJ, Parhar IS. Citalopram (antidepressant) administration causes sexual dysfunction in male mice through RF-amide related peptide in the dorsomedial hypothalamus. *Neuropharmacology* 2010;59:77–85.
- Steru L, Chermat R, Thierry B, Simon P. The tail suspension test: a new method for screening antidepressants in mice. *Psychopharmacology (Berl)* 1985;85:367–70.
- Steru L, Chermat R, Thierry B, Mico JA, Lenegre A, Steru M, et al. The automated tail suspension Test: a computerized device which differentiates psychotropic drugs. *Prog Neuropsychopharmacol Biol Psychiatry* 1987;11:659–71.
- Wei XH, Cheng XM, Shen JS, Wang ZT. Antidepressant effect of Yueju-Wan ethanol extract and its fractions in mice models of despair. *J Ethnopharmacol* 2008;117:339–44.
- Williams SM, Bryan-Lluka LJ, Pow DV. Quantitative analysis of immunolabeling for serotonin and for glutamate transporters after administration of imipramine and citalopram. *Brain Res* 2005;1042:224–32.
- Willner P, Muscat R, Papp M. Chronic mild stress-induced anhedonia: a realistic animal model of depression. *Neurosci Biobehav Rev* 1992;16:525–34.
- Witte S, Loew D, Gaus W. Meta-analysis of the efficacy of the acetonic kava-kava extract WS1490 in patients with non-psychotic anxiety disorders. *Phytother Res* 2005;19:183–8.
- Zhu W, Ma S, Qu R, Kang D. Antidepressant-like effect of saponins extracted from Chaihu-jia-longgu-muli-tang and its possible mechanism. *Life Sci* 2006;79:749–56.